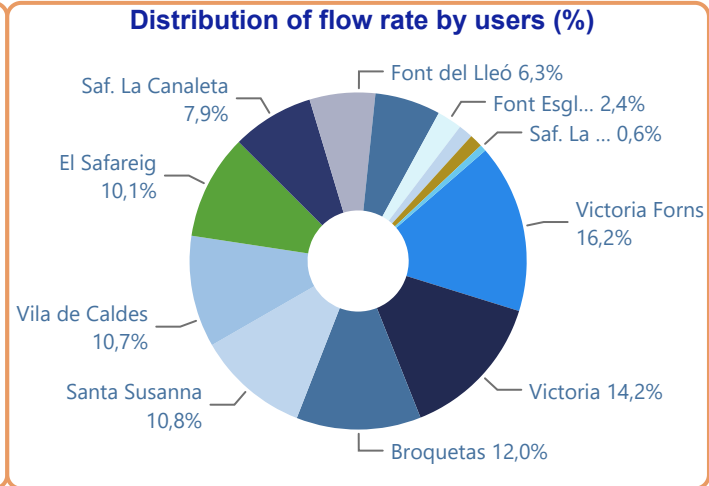
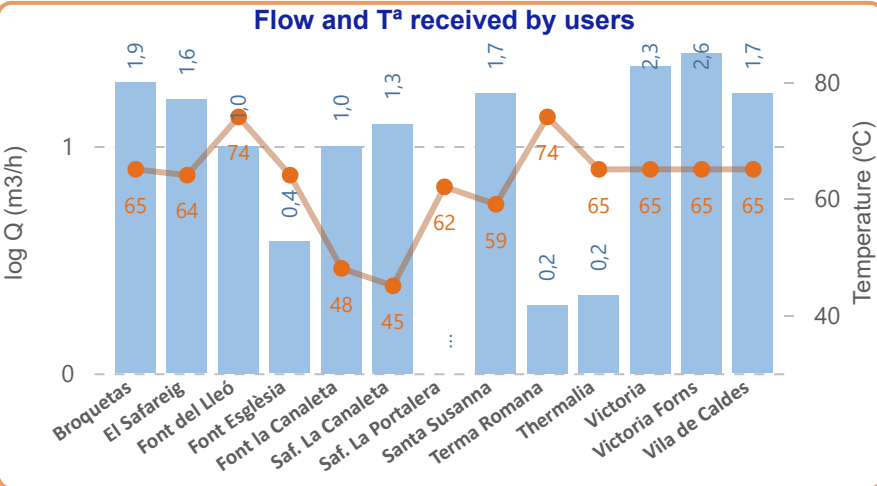


Physical Thermal Territory (PTT):

Caldes de Montbui

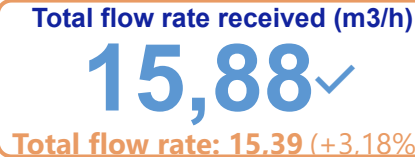
Thermal Resources and Users

This section presents the characteristics of the geothermal resource, including thermal water flow rates, source temperatures, distribution among current users, and the overall use of the thermal resource. It also provides an estimate of the gross geothermal energy available from the thermal springs before considering operational constraints or energy recovery processes.

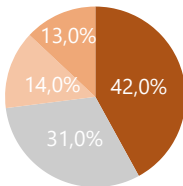


Thermal Water Types

- EFTh: Cold Thermal Water;
- ECTh: Hot Thermal Water;
- ETTh: Warm Thermal Water



Breakdown of annual thermal water consumption



- ECTh Spa
- ECTh Not used
- ECTh Springs
- ECTh Other

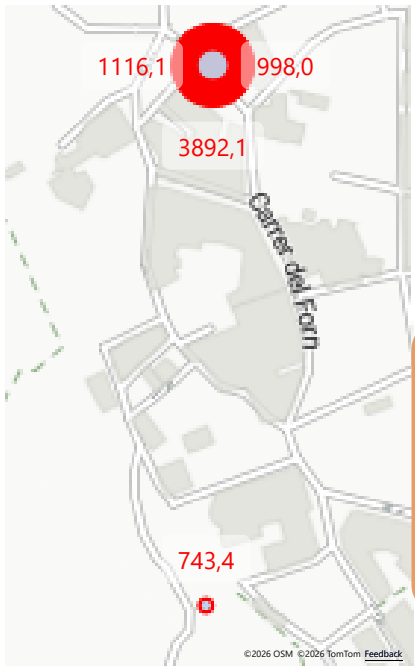
Total Geothermal Potential

0.95 MW

• Theoretical thermal power available from the geothermal resource, calculated from the source flow rate and the temperature difference between the spring outlet temperature and a reference temperature of 20°C.

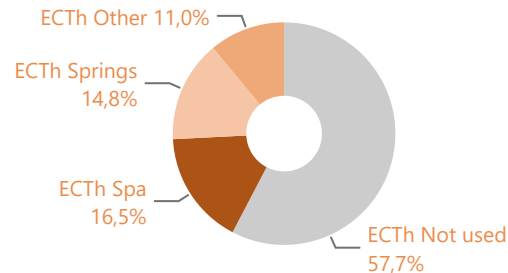
Water Heat Discharge

This section identifies and quantifies the main waste heat sources generated throughout the thermal water process. It presents the available recoverable heat resource from unused thermal water, thermal wastewater and existing cooling processes, highlighting the potential for future energy recovery and valorization.



Use	Flow (m³/h)	T inlet (°C)	T output (°C)	Heat Energy (MWh/year)
ECTh Spa	13,70	74	66	1116,1
ECTh Not used	7,80	74	25	3892,1
ECTh Other	7,30	35	25	743,4
ECTh Springs	2,00	74	25	998,0

Distribution of waste heat recovery potential



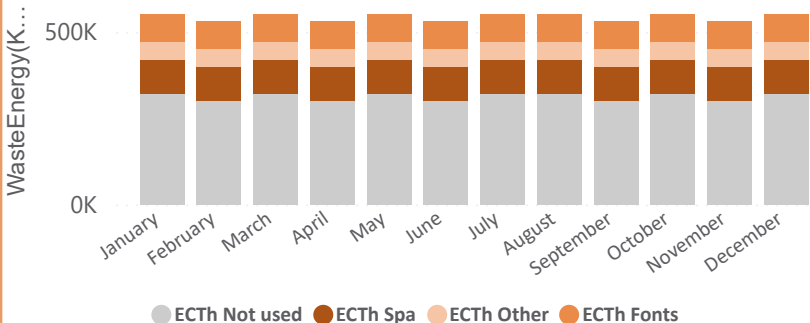
Recoverable heat energy (MWh th/year)

Total waste heat potential

6749,6✓

• Recoverable Heat Energy: the maximum recoverable thermal energy contained in wastewater after use, calculated from the temperature reduction between the supplied water on the discharge water.

Monthly distribution of recoverable waste heat sources



Actual Energy Uses

User	Domestic hot water	Heating	Watering
Ajuntament		X	
Balneari Broquetas	X	X	
El Safareig	X	X	
Font de l'Església			X
Safareig Portaleria			X
Santa Susana	X	X	
Spa Vila de Caldes	X		
Termes Victoria	X		

Potential Energy Uses

PotentialWasteWaterUse	Viable
Cooling	X
District heating network	X
Greenhouse heating	X
Mechanical Energy	X
Power Generation	X
Space & DHW heating	X
Thermal energy storage	X